# HeightMapNet: Explicit Height Modeling for End-to-End HD Map Learning

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- Problem/Objective
  - (Height aware) HD map construction

- Contribution/Key Idea
  - Novel enhanced view-transform module
  - PV-BEV transform 에서 height를 고려
  - Self-supervised를 통한 Foreground-Background 분리
  - Multi-scale feature fusion mechanism

김범준

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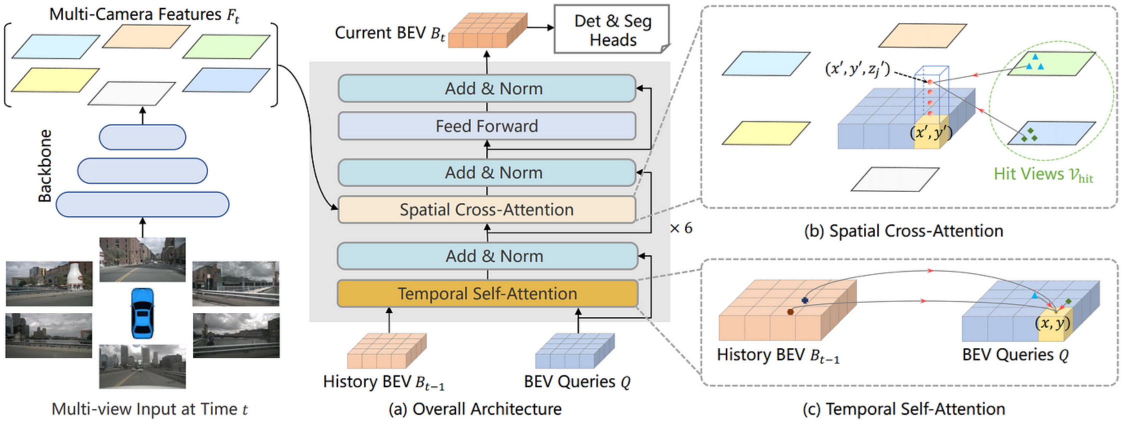
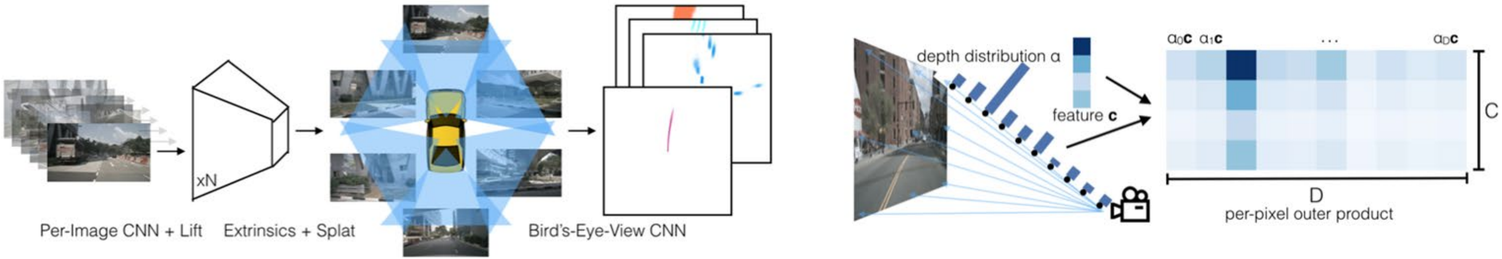



BEVFormer




Lift - Splat - Shoot

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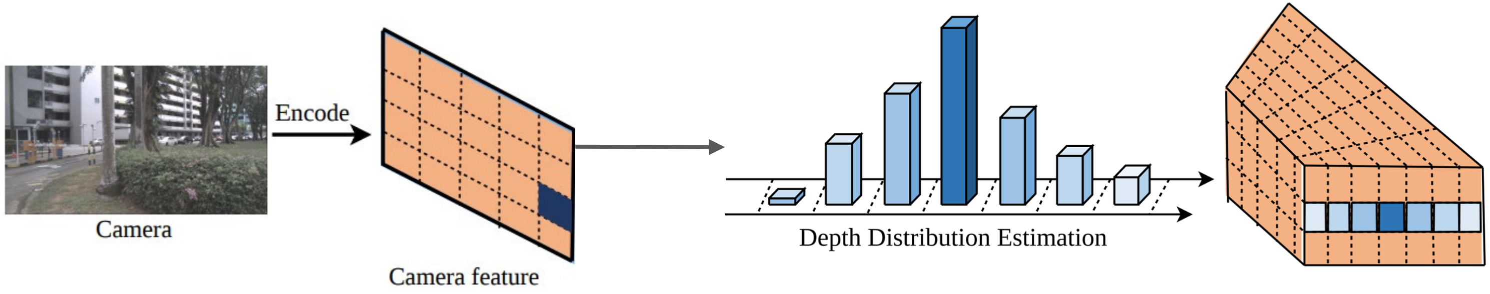


Lift - Splat 네트워크

- BEV로의 변환 과정
  - depth estimation → 3d frustum → (flatten) → BEV 변환
- 기존의 depth estimation 방법 or Query-based attnetion 방법은 주행환경의 높낮이에 관심이 x
  - 내리막길, 오르막길에서 성능 저하 → depth가 아닌 height estimation을 해보겠다.
  - 카메라 설치 높이에 대한 민감도를 감소시켜보겠다.
- Non-critical elements(하늘, 불필요한 배경)을 효과적으로 필터링 x
  - det분야처럼 이런 것도 필터링 해보겠다.

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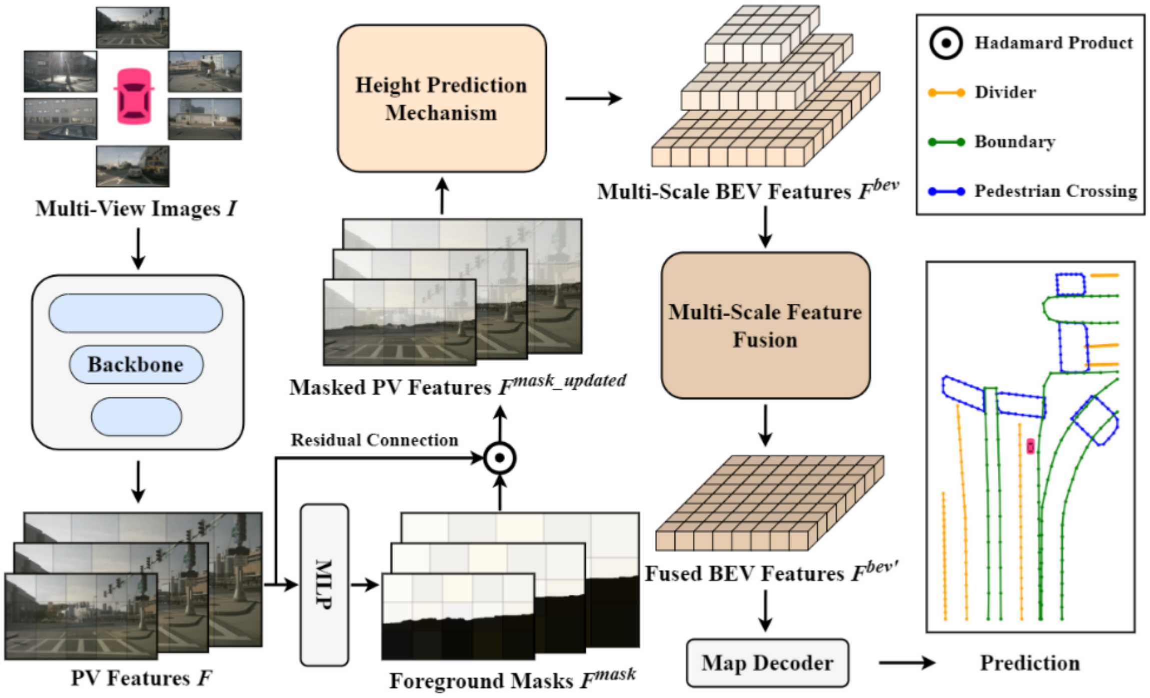


Figure 1. The overview of our proposed HeightMapNet. Our method consists of three main components: a foreground-background separation network that emphasizes roadway features, a height prediction mechanism for PV-to-BEV transformation, and a multi-scale feature fusion module that enhances BEV representation.

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## [HeightMapNet: Explicit Height Modeling for End-to-End HD Map Learning](#)

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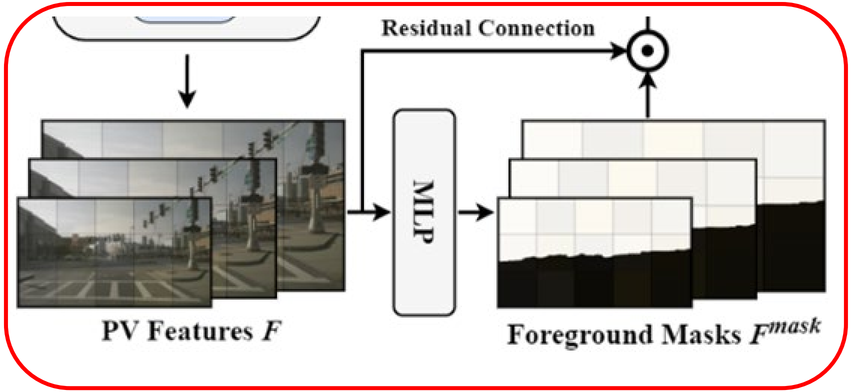


multi-scale PV features $F_i$

$$F_i^{mask} = \text{Sigmoid}(\text{Conv}(\text{ReLU}(\text{Conv}(F_i)))),$$
$$F_i^{mask\_updated} = F_i + F_i \odot F_i^{mask},\ i = 1, \dots, s.$$

- Self-supervised 를 활용
  - BEV 상의 reference point [-2.0m, 2.0m]
  - 3D → 2D 로 projected (by Camera intrinsic/extrinsic)
  - Projected 된 픽셀은 foreground로 분류
  - Projected 되지 않은 픽셀은 background로 분류

$$L_{mask} = \sum_i D_{\text{Manhattan}}(F_i^{mask}, F_i^{mask\_GT}),$$

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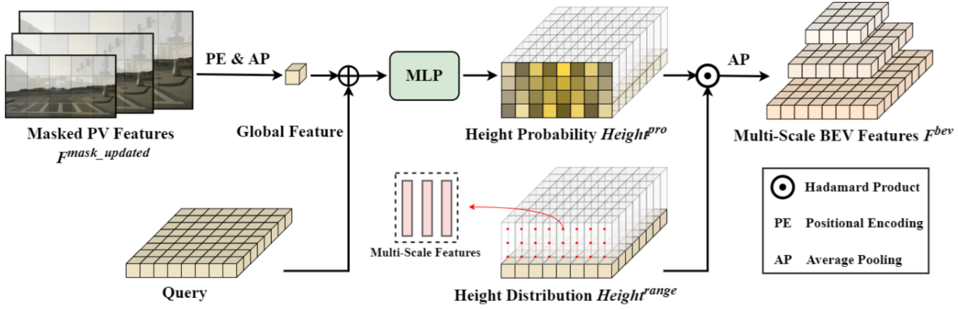


Figure 2. Illustration of height prediction mechanism.

$$Height^{pro} = \mathrm{MLP}(\mathbb{AP}(F_i^{mask\_updated} + \mathbb{PE}) + Query)$$

Height distribution probability

학습가능한 파라미터

intrinsic / extrinsic

$$Height_i^{range} = \mathrm{Sampling}_i(\mathbb{I} \cdot \mathbb{K} \cdot P_{3d}^{bev}),$$

- BEV reference point를 PV에 투영
- 3D → 2D 투영하고, sampling으로 PV feature 추출

$$F_i^{bev} = \mathbb{AP}(Height^{pro} \odot Height_i^{range}),$$

- 곱셈은 각 높이에 대해 PV 특징을 **확률적으로 가중**

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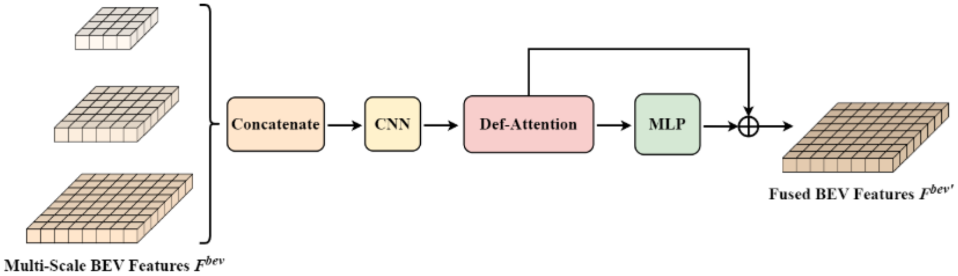


Figure 3. Illustration of multi-scale feature fusion.

$$
\begin{aligned}
F^{bev'} = & \mathrm{MLP}(\mathbb{DA}(\mathrm{Conv}(\mathrm{Concat}(F_1^{bev}, \ldots, F_s^{bev})))) \\
& + \mathbb{DA}(\mathrm{Conv}(\mathrm{Concat}(F_1^{bev}, \ldots, F_s^{bev}))).
\end{aligned}
$$

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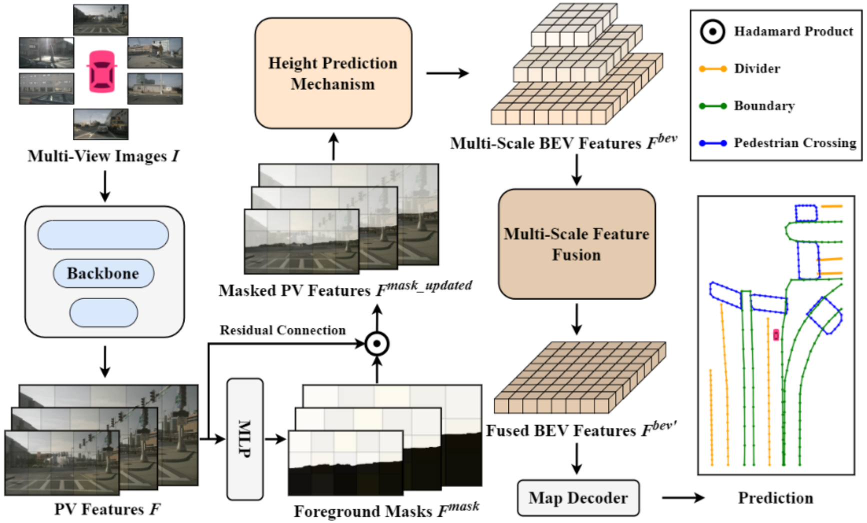


Figure 1. The overview of our proposed HeightMapNet. Our method consists of three main components: a foreground-background separation network that emphasizes roadway features, a height prediction mechanism for PV-to-BEV transformation, and a multi-scale feature fusion module that enhances BEV representation.

$$L = \lambda L_{cls} + \alpha L_{pos} + \beta L_{dir} + \gamma L_{mask}.$$

four pivotal components of loss: classification loss $L_{cls}$, point-to-point loss $L_{pos}$, edge direction loss $L_{dir}$, and mask loss $L_{mask}$, represented as follows:

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Table 1. Comparisons on the nuScenes `val` set. This table compares the performance of different sensors and architectures including Camera (C), Lidar (L), EfficientNet-B0 [24] (EB0), PointPillars [8] (PPs), and ResNet50 [6] (R50). † means the result is reproduced with public code, and ∗ indicates that HeightMapNet is extended by incorporating the one-to-many loss and BEV loss from MapTRv2.

| Methods | Modality | Backbone | Epochs | $AP_{div.}\uparrow$ | $AP_{bou.}\uparrow$ | $AP_{ped.}\uparrow$ | $mAP\uparrow$ |
|---|---|---|---|---|---|---|---|
| HDMapNet [9] | C | EB0 | 30 | 0.217 | 0.330 | 0.144 | 0.230 |
| HDMapNet [9] | C & L | EB0 & PPs | 30 | 0.296 | 0.467 | 0.163 | 0.310 |
| VectorMapNet [17] | C | R50 | 110 | 0.473 | 0.393 | 0.361 | 0.409 |
| VectorMapNet [17] | C & L | R50 & PPs | 110 | 0.505 | 0.475 | 0.376 | 0.452 |
| MapTR† [12] | C | R50 | 24 | 0.512† | 0.531† | 0.493† | 0.512† |
| PivotNet [5] | C | R50 | 24 | 0.565 | 0.601 | 0.562 | 0.576 |
| MapVR [35] | C | R50 | 110 | 0.618 | 0.594 | 0.550 | 0.588 |
| MapTRv2† [13] | C | R50 | 24 | 0.608† | 0.616† | 0.566† | 0.597† |
| HeightMapNet | C | R50 | 24 | **0.628** | 0.604 | 0.543 | 0.591 |
| HeightMapNet* | C | R50 | 24 | 0.626 | **0.622** | **0.604** | **0.617** |

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Table 3. Comparisons on the nuScenes `test` set. The input modality utilized is Camera. † means the result is reproduced with public code, and ∗ indicates that HeightMapNet is extended by incorporating the one-to-many loss and BEV loss from MapTRv2. All the results are obtained on a same device, with a training duration of 24 epochs and ResNet50 serving as the backbone.

| Methods | *General CD*: $\{0.5, 1.0, 1.5\}m\uparrow$ | | | | *Tighter CD*: $\{0.2, 0.5, 1.0\}m\uparrow$ | | | |
|---|---|---|---|---|---|---|---|---|
| | $AP_{div.}$ | $AP_{bou.}$ | $AP_{ped.}$ | $mAP$ | $AP_{div.}$ | $AP_{bou.}$ | $AP_{ped.}$ | $mAP$ |
| MapTR† [12] | 0.682† | 0.619† | 0.617† | 0.639† | 0.449† | 0.359† | 0.339† | 0.382† |
| MapTRv2† [13] | 0.762† | 0.700† | 0.690† | 0.717† | 0.544† | 0.425† | 0.401† | 0.457† |
| HeightMapNet | **0.773** | 0.693 | 0.665 | 0.710 | 0.552 | 0.421 | 0.369 | 0.447 |
| HeightMapNet* | 0.772 | **0.706** | **0.721** | **0.733** | **0.565** | **0.432** | **0.422** | **0.473** |

$$d_{\text{Chamfer},A\to B} = \frac{1}{|A|}\sum_{a\in A}\min_{b\in B}\|a-b\|^2$$

$$d_{\text{Chamfer},B\to A} = \frac{1}{|B|}\sum_{b\in B}\min_{a\in A}\|b-a\|^2$$

$$D_{\text{Chamfer}}(A,B) = d_{\text{Chamfer},A\to B} + d_{\text{Chamfer},B\to A}$$

- 더 엄격한 임계값({0.2, 0.5, 1.0}m) 하에서도 높은 정확도를 유지

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Table 4. Comparisons on the Argoverse 2 `val` set. The input modality utilized is Camera. † means the result is reproduced with public code, and $*$ indicates that HeightMapNet is extended by incorporating the one-to-many loss and BEV loss from MapTRv2. "-" indicates that the corresponding results are not available.

| Methods | Epochs | $AP_{div.}\uparrow$ | $AP_{bou.}\uparrow$ | $AP_{ped.}\uparrow$ | $mAP\uparrow$ |
|---|---|---|---|---|---|
| HDMapNet [9] | 6 | 0.057 | 0.376 | 0.131 | 0.188 |
| VectorMapNet [17] | 24 | 0.361 | 0.392 | 0.383 | 0.379 |
| MapTR† [12] | 6 | 0.533† | 0.562† | 0.509† | 0.535† |
| MapVR [35] | - | 0.600 | 0.580 | 0.546 | 0.575 |
| MapTRv2† [13] | 6 | 0.704† | 0.664† | 0.620† | 0.663† |
| HeightMapNet | 6 | 0.639 | 0.627 | 0.667 | 0.644 |
| HeightMapNet* | 6 | **0.706** | **0.679** | **0.627** | **0.671** |

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Table 5. Ablation study on the nuScenes `val` set. This ablation study details the incremental addition of various modules to the baseline. Auxiliary Loss in configuration #5 indicates that the one-to-many loss and BEV loss from MapTRv2 have been added to the basic HeightMapNet.

| # | Methods | $AP_{div.}\uparrow$ | $AP_{bou.}\uparrow$ | $AP_{ped.}\uparrow$ | $mAP\uparrow$ |
|---|---|---|---|---|---|
| 1 | Baseline (MapTR) | 0.512 | 0.531 | 0.493 | 0.512 |
| 2 | + Height Prediction Mechanism | 0.571 | 0.566 | 0.516 | 0.551 |
| 3 | + Foreground-Background Separation | 0.570 | 0.567 | 0.555 | 0.564 |
| 4 | + Multi-Scale Feature Fusion (HeightMapNet) | **0.628** | 0.604 | 0.543 | 0.591 |
| 5 | + Auxiliary Loss (HeightMapNet*) | 0.626 | **0.622** | **0.604** | **0.617** |

Table 6. Ablation study on different height values. This ablation study delineates the impact of varying the number of road surface height sampling points on the performance of HeightMapNet.

| Height Values | $AP_{div.}\uparrow$ | $AP_{bou.}\uparrow$ | $AP_{ped.}\uparrow$ | $mAP\uparrow$ |
|---|---|---|---|---|
| 8 | 0.619 | 0.593 | **0.552** | 0.588 |
| 12 | **0.628** | **0.604** | 0.543 | **0.591** |
| 16 | 0.618 | 0.600 | 0.549 | 0.589 |

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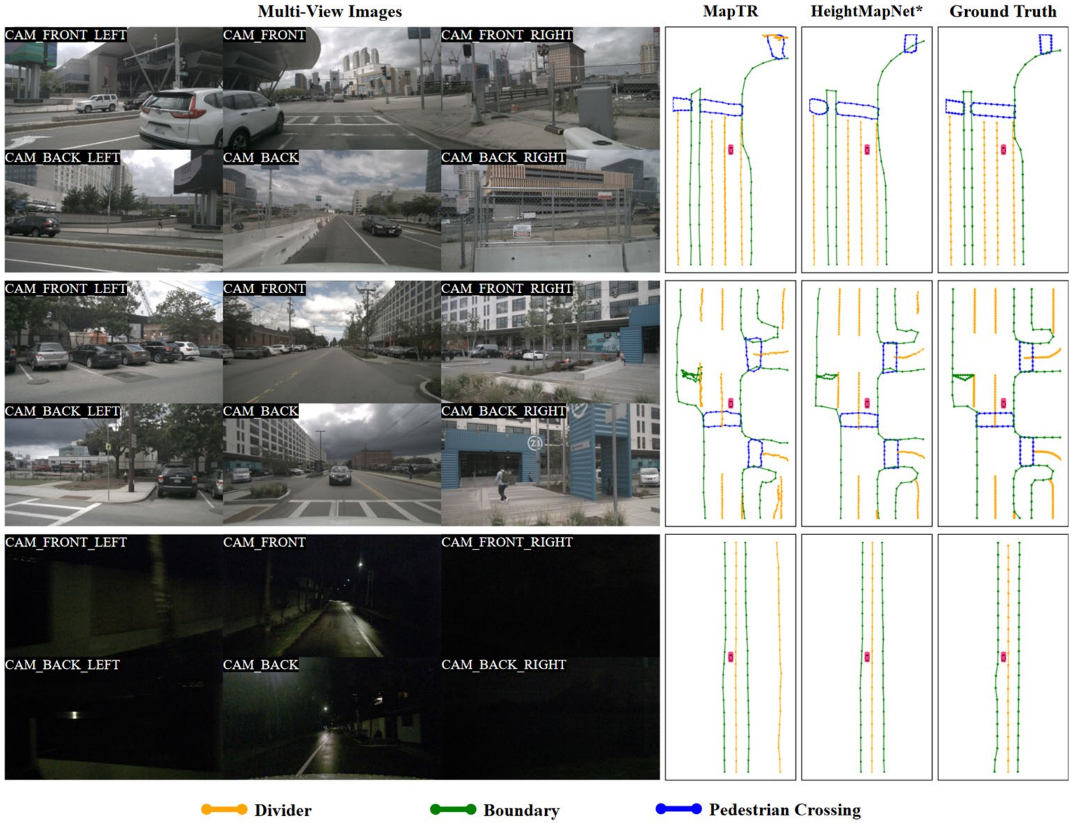


Figure 4. Visualizations on the nuScenes `val` set under different weather conditions.